SKILL BUILDERS · UNIT 8 · SCORE 3

Transformations: Practice Set

NC.8.G.2–4

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.G.3

Translate left 2 and down 3: the point **(2, 5)**. Where does it land?

(A) (−1, 3)
(B) (0, 2)
(C) (4, 8)
(D) (0, 8)

2 Score 3 NC.8.G.3

Reflect over the y-axis: the point **(−3, 4)**. Where does it land?

(A) (4, −3)
(B) (−3, −4)
(C) (3, −4)
(D) (3, 4)

3 Score 3 NC.8.G.3

Reflect over the x-axis: the point **(6, −1)**. Where does it land?

(A) (6, 1)
(B) (−6, 1)
(C) (−1, 6)
(D) (−6, −1)

4 Score 3 NC.8.G.3

Rotate 180° about the origin: the point **(2, 7)**. Where does it land?

(A) (−2, −7)
(B) (−2, 7)
(C) (−7, 2)
(D) (7, 2)

5 Score 3 NC.8.G.4

Dilate by k = 4: the point **(1, 3)**. Where does it land?

(A) (4, 3)
(B) (4, 12)
(C) (5, 7)
(D) (0.25, 0.75)

6 Score 3 NC.8.G.3

Rotate 90° counterclockwise about the origin: the point **(5, 5)**. Where does it land?

(A) (−5, −5)
(B) (5, −5)
(C) (5, 5)
(D) (−5, 5)

7 Score 3 NC.8.G.4

Dilate by k = 1/2: the point **(4, −6)**. Where does it land?

Ⓐ (4.5, −5.5)

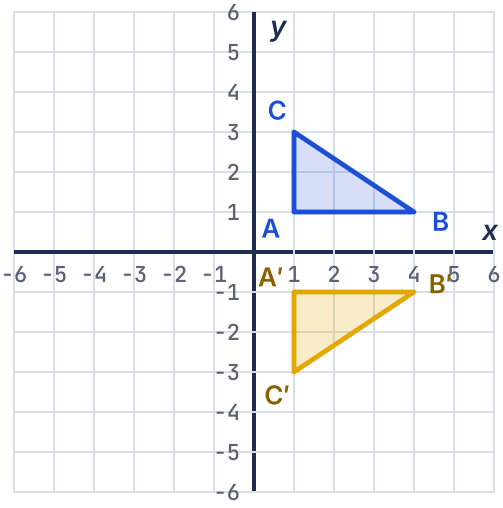
Ⓑ (2, −3)

Ⓒ (2, 3)

Ⓓ (8, −12)

8 Score 3 NC.8.G.3

Which move takes the blue triangle to the gold triangle?



Ⓐ Translation down 4

Ⓑ Reflection over the x-axis

Ⓒ Dilation by k = 2

Ⓓ Rotation 180°

9 Score 3 NC.8.G.2

After a translation, a reflection and a rotation, the image is...

Ⓐ congruent: same size and shape

Ⓑ similar but smaller

Ⓒ similar but bigger

10 Score 3 NC.8.G.3

First translate right 3, then reflect over the x-axis: the point **(1, 2)**. Where does it land?

Ⓐ (4, 2)

Ⓑ (4, −2)

Ⓒ (−4, −2)

Ⓓ (1, −5)

11 Score 3 NC.8.G.3

Find the mistake. Mia reflected (2, 6) over the x-axis and got **`(-2, 6)`**. What went wrong?

Ⓐ Mia changed x. Over the x-axis, y changes sign: (2, −6).

Ⓑ Mia is right.

Ⓒ The numbers should switch: (6, 2).

Ⓓ Both signs should change: (−2, −6).